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Editor-in-Chief
*IEEE Access*

**Subject: Submission of the manuscript "Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization"**

Dear Editor,

We are pleased to submit the above manuscript for consideration in *IEEE Access*. The work is a complete rebuild of a manuscript that was previously reviewed by the journal and rejected under the title "Coordinated V2G Voltage and Frequency-State Regulation in a Two-Area System: A Gaussian Actor–Critic Simulation Framework" (manuscript ID: [previous ID]). The reviewers of that version found, correctly, that the controller had not been trained and that the network and frequency models were algebraic surrogates. Rather than revise the text, we rebuilt the study from the model upward and are submitting it as a new manuscript. A point-by-point response to the earlier reviews is enclosed so that the Editor can see how each comment was addressed.

**What the paper does.** An aggregated fleet of vehicle-to-grid electric vehicles can support distribution voltages and provide frequency reserve, but both services draw on the same converter rating and stored energy. The paper formulates their coordination as a two-time-scale control problem: every hour each area aggregator chooses a fleet power set-point and the share of its plugged-in converter rating held as frequency reserve. The set-points are evaluated by AC power flow on IEEE 33-bus feeders in two areas with an energy-consistent fleet model; the reserve sets the gain and limit of a V2G droop loop in a dynamic two-area load-frequency control model with inertia, governors, a tie line, and automatic generation control, simulated at a 20-ms step. A proximal policy optimization agent is trained with 5 independent seeds and compared, under identical conditions, with uncontrolled charging, a rule-based volt-watt and droop controller, and a day-ahead optimization with perfect foresight.

**Main findings.** On 30 held-out days the learned policy lowers the daily performance cost by 11.0%, 44.8%, and 71.6% relative to the three benchmarks, reduces undervoltage by 13.5% and the mean frequency nadir by 5.0% relative to the day-ahead optimization, and keeps its advantage when that benchmark is re-optimized for every reserve share and even against a per-day oracle choice of the share. The paper also reports what the method does not achieve: some hours of the weak residential feeder stay below 0.95 p.u. with every controller, a downward activation of the committed reserve can push further hours below the band, and consecutive-day operation shows a downward drift of the residential fleet energy that a soft terminal penalty does not prevent. These limits are quantified in the manuscript rather than left to the reader.

**Why the work suits *IEEE Access*.** The paper combines power-system modeling (verified AC power flow and a dynamic LFC model in physical units), reinforcement learning with a stated training and evaluation protocol, and an unusually complete reproducibility package. The energy exchanged by the droop loop is included in the fleet energy balance and checked against exact 3600-s simulations; every number in the manuscript is recomputed from the raw result files by an independent verification script (84 checks); and the code, trained policies, result files, checksums, and exact package versions are released at `https://github.com/kenoto99/v2g-ppo-two-area` (archived on Zenodo, DOI 10.5281/zenodo.23851585).

**Declarations.** The manuscript is original, has not been published, and is not under consideration elsewhere. All authors have read and approved the submitted version and agree to its submission to *IEEE Access*. The author list differs from the previously reviewed version: one author withdrew and one author

was added, with the agreement of all authors. The authors declare no conflict of interest. No human or animal subjects were involved. The work was supported in part by Rajamangala University of Technology Thanyaburi.

We would be grateful if the manuscript could be handled by an Associate Editor in the area of power and energy systems or smart grids. Thank you for considering our submission.

Sincerely,

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